

The empirical regulatory operator of CD4⁺ T cells

A low-dimensional, condition-gated, partly cell-type-specific reading of a genome-scale Perturb-seq screen

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Abstract

Genome-scale Perturb-seq screens are typically read as ranked lists of individual regulator hits. We argue that, in primary human CD4⁺ T cells, the perturbation effects instead form a shared **regulatory operator** that can be estimated, interrogated geometrically, and used to predict — and that treating it as one object, rather than a hit list, recovers biology the ranked view cannot. Working from a genome-scale CRISPRi Perturb-seq atlas (four donors $\times$ three activation states; ~ 1.8 TB raw) entirely on a laptop, we assemble a regulator $\times$ gene $\times$ condition effect tensor in a power-decoupled z-score space (confound $\rho = -0.006$) and establish three results. First, the operator’s **predictive subspace is low-dimensional (~ 7 components)**: a low-rank fit predicts the unobserved late-stimulation response of out-of-panel regulators, beating a persistence baseline at every rank across a $4\times$ expansion of the regulator axis (3,106 regulators; aggregate margin $\Delta R^2 = +0.379$). The advantage survives strength stratification — even the strongest, panel-matched regulators, for which persistence is a fair baseline, stay positive ($+0.21$) — so it is not an artifact of weak-regulator baseline collapse. This converts the operator from descriptive to predictive, with the honest caveat that the absolute precision is modest (model $R^2 \approx 0.06$ – 0.08) and softens at scale. Second, a CANDECOMP/PARAFAC decomposition resolves **gene programs gated by activation state** (bootstrap-CI-supported factors peaking in resting versus early-stimulated cells), assigned to regulators blind to annotation. Third, cross-cell-type concordance against an external K562 screen **separates universal from T-cell-specific regulation**: chromatin/SAGA machinery is universal (anchor SUPT20H), whereas the immune Rac $\rightarrow$ WAVE $\rightarrow$ actin cytoskeletal module (anchor DOCK2) is T-cell-specific and donor-reproducible — an independent screen recovering the same axis the decomposition named. Every claim is gated by an explicit control for the power–magnitude confound, cis-off-target artifacts, donor reproducibility, and decomposition stability; caveats are reported in the direction that costs signal. The contribution is not a new estimator or a new biological hit, but the operator view itself — inheriting calibrated differential-expression front-ends and adding low-rank predictivity, condition-resolved programs, and cell-type specificity on top.

1. Introduction

Genome-scale Perturb-seq — pooled CRISPR perturbation with single-cell transcriptomic read-out — has made it possible to measure the transcriptional consequence of knocking down essentially every expressed gene in a cell^{1,2}. In primary human CD4⁺ T cells, a recent screen extends this to a setting that matters for immunology and for therapy: four donors, three activation

states (resting, early- and late-stimulated), and genome-scale coverage of candidate regulators. The standard way such a screen is read is as a ranked list — which perturbations move the most genes, which are the “hub” regulators — and that reading is useful but incomplete. It treats each regulator’s effect as an independent entry and discards the possibility that the effects share structure: that the response to one perturbation is informative about the response to another.

Here we take the opposite starting point. We treat the screen as a single object — the regulator $\times$ gene $\times$ condition tensor of perturbation effects, which we call the **regulatory operator** — and ask what its geometry reveals. This is a different question from ranking perturbations or extracting gene programs, and to our knowledge it has not been asked of a primary-cell Perturb-seq screen: existing analyses either rank regulators by effect breadth or factor the expression matrix into programs, but none builds the regulator $\times$ gene operator and interrogates it as an operator — its predictive structure, its low-dimensional subspace, its decomposition by condition. That is the gap this work addresses, and the three results that follow are three properties of the operator: it predicts, it is gated by condition, and it separates universal from cell-type-specific regulation.

Positioning this against the single-cell-CRISPR methodology literature clarifies what is inherited and what is contributed. The front end — turning raw counts into calibrated, harmonized per-regulator effect estimates — is a solved problem we build on rather than redo: calibrated single-cell-CRISPR testing³ and the scPerturb effect-quantification framework⁴ define that layer. Our contribution sits above it: the operator, its condition factors, and its out-of-panel predictivity. The closest methodological neighbor is guided sparse factor analysis⁵, which recovers gene programs and their driving perturbations probabilistically; our CP/SVD operator decomposition is a deterministic analogue of that object, and we treat it as a comparison rather than a claim of priority. The predictive result is framed against a frontier where deep perturbation-response predictors do not yet reliably beat simple baselines^{6,7}: we therefore ask not whether a flexible model can fit, but whether the operator’s shared structure beats an honest persistence baseline out-of-panel — a lower, more falsifiable bar.

A methodological posture runs through all three results, and we state it as a choice rather than leave it implicit: fewer claims, each controlled for the confound that most threatens it. Every result in this paper is gated by an explicit control — for the power–magnitude confound that inflates naive effect sizes, for cis-off-target artifacts, for donor reproducibility, and for decomposition stability — and where a control costs signal, we report the reduced number. In a field where predictive claims often do not survive a fair baseline, this discipline is part of the argument, not decoration around it.

Finally, a note on structure. The thesis has three legs — the operator, its condition-gated programs, and the universal/T-cell-specific split — but the results open (Section 2, Figure 1) with neither: they open with the regulator ranking. This is deliberate. The operator is built on top of the ranking, and the ranking is only trustworthy if its effect metric is not an artifact of statistical power; Section 2 establishes exactly that — the ranking is metric-robust and power-controlled — so that the operator built on it in Sections 3–5 stands on an audited foundation rather than an assumed one. The ranking is the substrate, not a fourth claim; we establish it first and then build.

2. The regulator ranking is metric-robust and power-controlled

Before treating the screen as an operator, we establish that its most basic summary — which regulators have the broadest transcriptional effect — is trustworthy. This matters because the

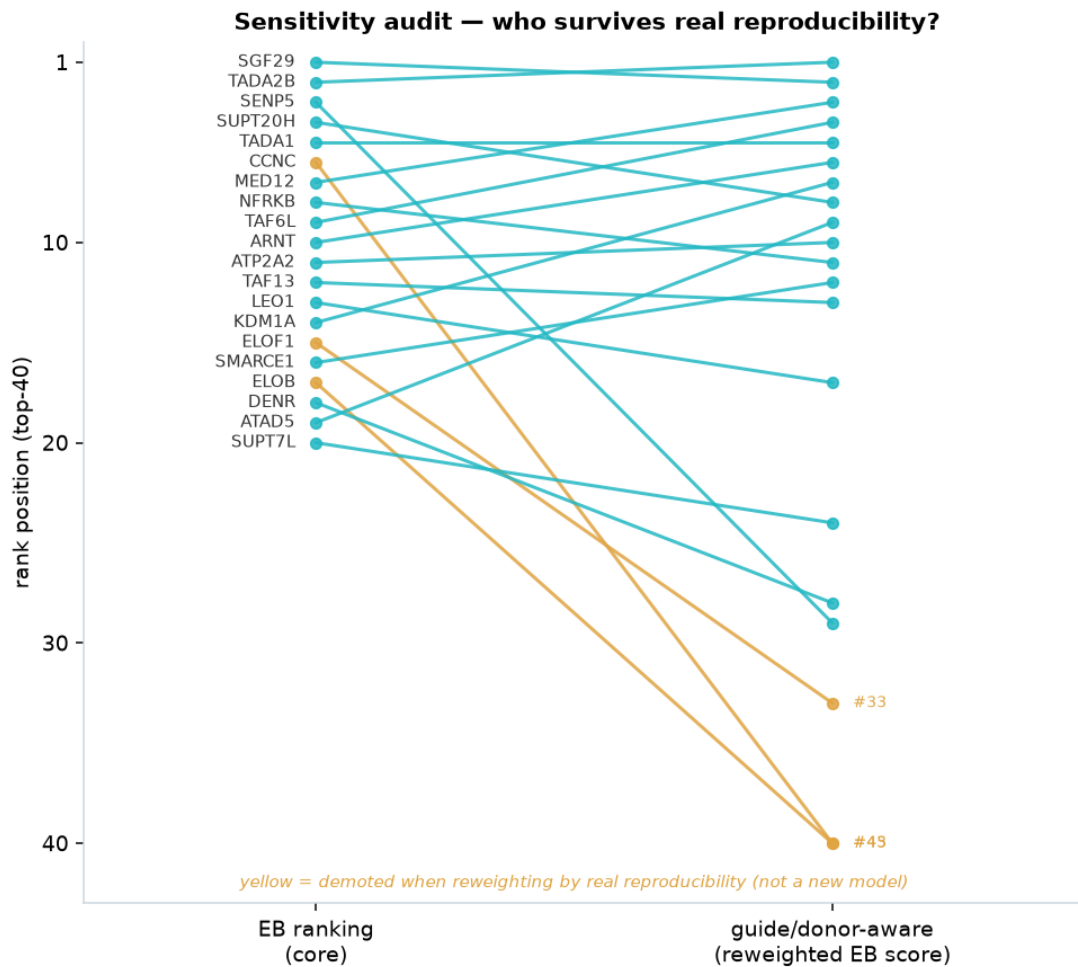


Figure 1. Donor reproducibility is carried as an annotation, not used to re-sort the ranking. Rank change induced when cross-donor reproducibility is folded into the breadth-based regulator ranking. Regulators that rank highly but fail cross-donor concordance — for example one carried by a single guide with no cross-guide check — stay visible at their rank, flagged rather than demoted.

operator is built on the same effect estimates the ranking uses; if those estimates were dominated by a statistical-power artifact, the operator would inherit it. This section audits the ranking and shows it survives the three challenges most likely to undermine it: the choice of effect metric, the confound between apparent effect and assay power, and cis-off-target contamination. It is the foundation the rest of the paper stands on, not a fourth scientific claim.

The ranking metric is not an artifact of the metric choice

A recurring criticism of Perturb-seq hub rankings is that they rank a *count* — how many genes pass a differential-expression threshold — rather than an effect size, so the ranking could be an artifact of the counting. We tested this directly by building a continuous effect magnitude (the summed absolute log-fold-change over the FDR-significant genes for each regulator) and comparing it to the count-based rank. The two agree closely: Spearman $\rho = 0.90$ between the proper FDR-restricted magnitude and `n_downstream` (source `REVIEW_RESPONSE.md`; per-gene 0.93). A naive all-gene magnitude — summing over every gene rather than the significant ones — is *worse*, not better, because it sums noise. The ranking is therefore robust to the metric: doing the effect-size calculation properly reproduces the count-based order, and the order does not depend on which of the two defensible metrics is used.

The ranking is controlled for statistical power

The most serious confound is power. Regulators assayed in more cells can show larger apparent effects for purely statistical reasons, so a ranking that simply correlated with cell count would be measuring assay depth, not biology. In raw log-fold-change space this confound is real and large: the per-regulator effect-vector norm correlates with cell count at Spearman $\rho = -0.68$ (the sign reflects the particular normalization, but the magnitude is the concern). The ranking metric we use does not carry it: the FDR-restricted magnitude correlates with cell count at only $\rho = -0.21$, and `n_downstream` even less. The power confound is a property of the naive magnitude, not of the ranking, and it is precisely why every downstream operator analysis is performed in the pooled z-score space (Section 7) where the confound is removed.

Two further properties make the ranked signal interpretable. Effective knockdown gates it: 62% of contrasts have a statistically significant on-target knockdown, and those concentrate 85% of all trans-effects — so the broad-effect regulators are overwhelmingly the ones whose intended perturbation actually took hold. And the ranking is externally corroborated: the top regulators overlap an independent T-cell CRISPR screen at 25–30 of the top 100 ($p < 1e-26$), a concordance far above chance for two screens with different readouts.

Donor reproducibility is reported, not used to re-sort

The screen spans four donors, and cross-donor reproducibility is informative about which high-ranking regulators are robust. We carry donor-robustness as an annotation column on the ranking (`hub_ranking_bayes.csv`), not as a re-sorting criterion. This is deliberate: re-ranking on donor concordance would silently hide the regulators that rank high but fail reproducibility, which is exactly the population a reader needs to see. As a column, it lets a reader see “ranked highly, but fails cross-donor concordance” — for example a regulator carried by a single guide with no cross-guide check stays visible at its rank, flagged rather than demoted (Figure 1).

The ranking is robust to cis-off-target exclusion

CRISPRi can knock down not only the targeted gene but its chromosomal neighbors, so an apparent regulator could be acting through a cis neighbor rather than the annotated target. The screen’s `offtarget_flag` captures this, and we confirmed it is overwhelmingly a cis signal: it agrees with an independent neighboring-gene-knockdown call at 99.4%, of which the flagged cases are 92% cis. Excluding flagged regulators is therefore a legitimate specificity control. It does reshape parts of the ranking — the SAGA chromatin complex is the clearest case, where two subunits (TADA2B, TAF6L) prove cis-inflated and collapse under exclusion while the cis-clean subunit SUPT20H rises (rank 4 $\rightarrow$ 3). The complex’s identity survives on its cis-clean members, but the leading-subunit list is corrected; we carry this correction forward into the program naming of Section 4, where SUPT20H alone anchors the SAGA program.

Taken together, these audits establish the ranking as an audited substrate: metric-robust, power-controlled, externally corroborated, transparent about donor reproducibility, and cleaned of cis-off-target artifacts. The operator built on these effect estimates in the following sections therefore inherits a controlled foundation rather than an assumed one.

3. A shared regulatory operator with a low-dimensional predictive subspace

Having established that the regulator ranking is trustworthy (Section 2), we now treat the screen not as a ranked list but as a single object: the regulator $\times$ gene $\times$ condition tensor of perturbation effects, which we call the regulatory operator. The question this section answers is whether that object has *shared* structure — structure that lets the measured perturbations inform the unmeasured response of others — or whether each regulator’s effect profile is independent, in which case the operator is merely a stack of unrelated rows and the ranked-list view loses nothing.

Construction

We assemble the operator as a regulator $\times$ gene $\times$ condition tensor in a pooled z-score space, where each entry is a knockdown effect standardized at the layer level rather than per condition. This choice is load-bearing: raw log-fold-change magnitude is confounded with statistical power (regulators assayed in more cells show larger apparent effects; Section 2), and a tensor built in raw space would encode that confound as structure. In the pooled z-score space the confound is removed — the rank correlation between per-regulator slab norm and cell count is $\rho = -0.006$ on the panel genes the decomposition uses, and a power gate on the singular subspace holds at a maximum $|\rho|$ of 0.049. We build on the set of regulators above a per-condition cell-count floor, which yields 3,106 regulators — a fourfold expansion over the 800-regulator pilot on which the approach was first validated. The confound guard is re-asserted after the expanded fetch and passes, so the fourfold-larger operator is estimated in the same clean representation as the pilot.

The operator predicts the unobserved response of out-of-panel regulators

The central claim is predictive. We hold out an entire condition fiber — the late-stimulation (Stim48hr) response — for regulators that lie outside the original variance-selected panel, and ask whether a low-rank fit trained on the other regulators, reading only the resting and early-stimulation entries of the held-out regulators, recovers their held-out late-stimulation response. The baseline is persistence: predicting that the late-stimulation response equals the early-

stimulation one. Persistence is not a straw man — if late stimulation simply extended early stimulation, it would be hard to beat; its negative R^2 (-0.305 on the full held-out set) shows that late stimulation genuinely diverges from early, so any method that beats it is using cross-regulator structure to anticipate that divergence.

On 621 held-out out-of-panel regulators, the low-rank model beats persistence **at every rank**, with an aggregate margin of $\Delta R^2 = +0.379$ at rank 7 (Figure 2). Beating the baseline at every rank, rather than at a single tuned rank, is what makes the result a property of the operator rather than of a lucky hyperparameter.

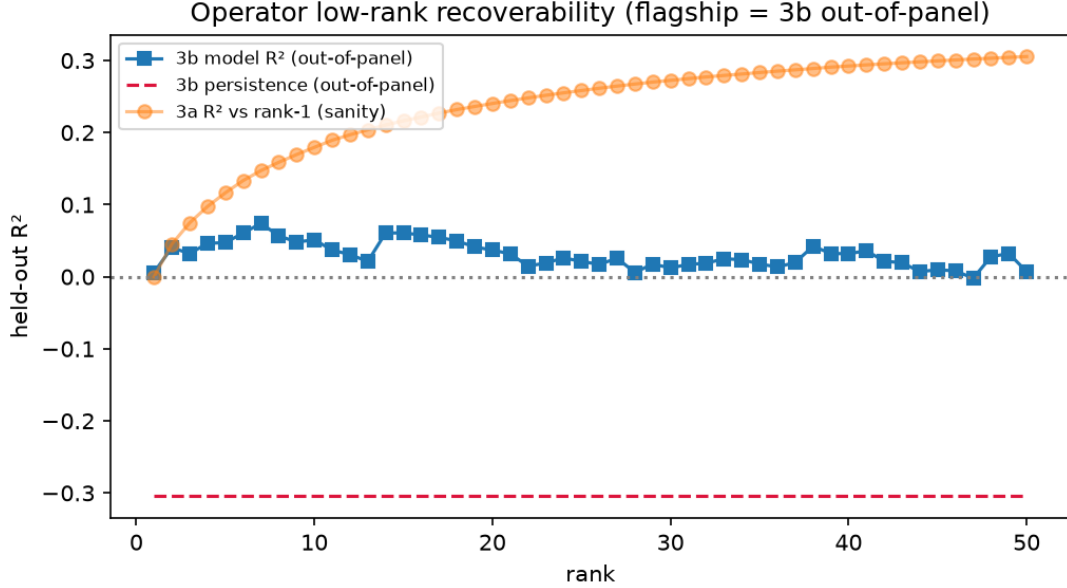


Figure 2. The operator’s predictive subspace is low-dimensional and beats persistence at every rank. Out-of-panel completion R^2 for the low-rank fit versus the persistence baseline across ranks, on 621 held-out out-of-panel regulators of the 3,106-regulator operator. The low-rank model beats persistence at every rank; the aggregate margin peaks at rank 7 ($\Delta R^2 = +0.379$) and declines thereafter — a genuine turnover that scopes the predictive subspace to ~ 7 dimensions.

The size of that margin, however, must be read carefully, and stratifying by knockdown strength is what keeps it honest (Table 1; source `operator_completion_stratified_3106.csv`). The margin is *largest* for weak regulators ($+0.501$, $n=310$) and *smallest* for strong ones ($+0.277$ on a median split, $n=311$; $+0.205$ on the subset matched to the original panel’s strength, $n=203$). This ordering is mechanistic, not mysterious: persistence is a fair baseline for strongly knocked-down regulators (its R^2 is only -0.146 in the panel-matched stratum) and collapses for weakly knocked-down ones (-0.421), so the aggregate margin is, if anything, concentrated where the baseline is weakest. The conservative reading is therefore the one we adopt: even for the strongest, panel-matched regulators — where persistence is a genuinely fair baseline — the model still beats it, by $+0.205$ at rank 7 and by at least $+0.151$ at every rank. The out-of-panel predictive advantage is not an artifact of weak-regulator baseline collapse; it holds where the baseline is hardest to beat.

Table 1. The out-of-panel predictive advantage survives strength stratification. Held-out late-stimulation completion at rank 7 (the peak rank), by knockdown-strength stratum, read from `operator_completion_stratified_3106.csv`. The margin is largest for weak regulators, where the persistence baseline collapses, but stays clearly positive even for the strongest, panel-matched regulators — for which persistence is a fair baseline — so the advantage is not an artifact of weak-regulator baseline collapse.

Stratum	n	Model R^2	Persistence R^2	Margin ΔR^2
All out-of-panel	621	0.074	−0.305	+0.379
Strong (median split)	311	0.069	−0.208	+0.277
Weak (median split)	310	0.080	−0.421	+0.501
Strong, panel-matched	203	0.058	−0.146	+0.205

Two boundaries on this claim are worth stating explicitly, because both are places a reader could otherwise over-read the result. First, the held-out unit is a condition *fiber*, not a whole regulator: each held-out regulator is observed in the resting and early-stimulation conditions, and only its late-stimulation response is predicted. A purely low-rank model cannot predict a regulator with no observed entries at all, and we make no such claim — the result is extrapolation across a regulator’s conditions, not imputation of never-measured regulators. Second, the absolute predictive precision is modest: the model’s held-out R^2 is in the range 0.06–0.08 across all strata, well below the 0.154 the pilot reached on its smaller 160-regulator holdout. The larger, more heterogeneous 3,106-regulator population makes the shared low-rank structure a modestly worse per-regulator predictor in absolute terms, even at matched strength. The beats-persistence *advantage* is robust and reproducible across strata and ranks; the absolute precision is modest and does not match the pilot. We report both halves, because reporting only the first would be an inflated headline.

The predictive subspace is low-dimensional

The rank at which prediction is best is itself informative. The out-of-panel completion R^2 peaks at rank 7 and declines thereafter through rank 50 (Figure 2) — a genuine turnover, not a curve still climbing at the edge of the swept range. This licenses a low-dimensionality statement, but a carefully scoped one: it is the **predictive subspace** that is low-dimensional (~ 7 components), not the operator as a whole. The operator’s own effective rank, measured on its singular spectrum, is roughly an order of magnitude higher (Figure S1); the ~ 7 -dimensional structure is the part of it that generalizes to unseen regulators. The distinction matters — a reader who took “low-rank operator” at face value would expect a spectrum that collapses after seven components, and the spectrum does not. What is low-rank is the shared, transferable structure, and it is a property of this 3,106-regulator holdout configuration rather than an intrinsic constant of T-cell regulation.

The representation stays pooled at scale

A fourfold expansion of the regulator axis could in principle break the pooling that makes the operator one object rather than many: if the added regulators occupied their own disjoint subspace, the “shared” structure would be an average over unrelated blocks. A CANDECOMP/PARAFAC decomposition of the expanded operator rules this out. Two factors are gated by condition with bootstrap confidence intervals excluding a flat profile (peaking in resting and in early-stimulated cells respectively; Section 4), and two further factors are cleanly constitutive — a cleaner decomposition than the pilot, which recovered no clean constitutive

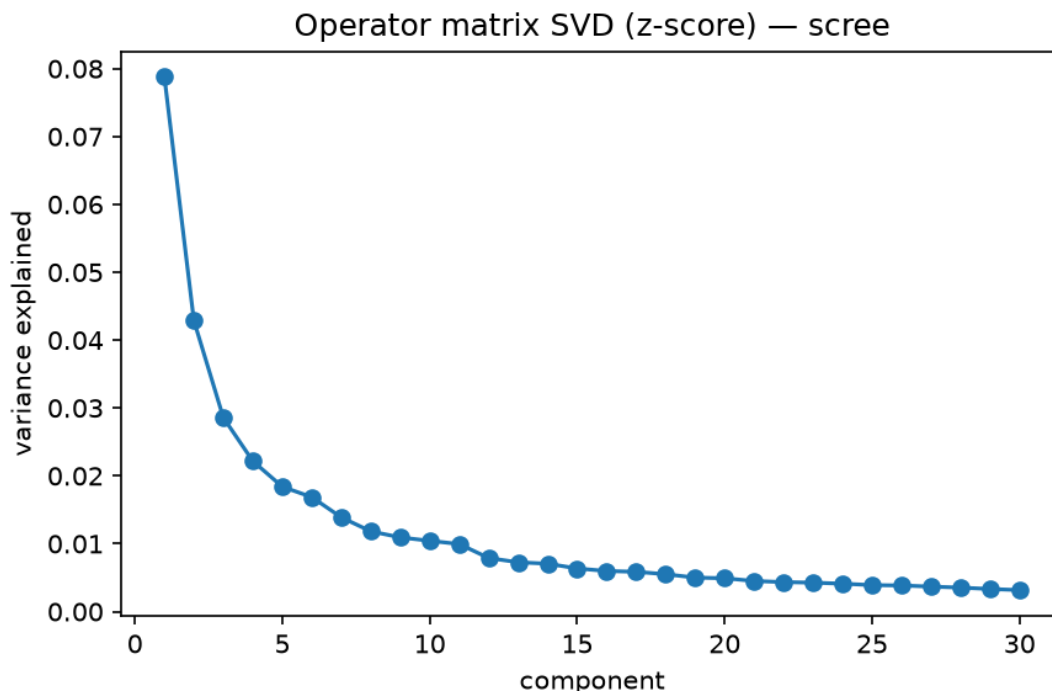


Figure S1. The operator’s own effective rank is far higher than its predictive subspace. Singular-value scree of the 3,106-regulator operator. The spectrum does not collapse after seven components (effective rank ≈ 86); the ~ 7 -dimensional structure identified in Figure 2 is the transferable part that generalizes to unseen regulators, not the operator’s full rank.

factor. That the decomposition finds coherent, condition-structured, partly-shared factors at 3,106 regulators is positive evidence that the representation remains genuinely pooled at the expanded scale, not a mosaic of independent per-regulator effects.

Taken together, these results convert the operator from a descriptive summary into a predictive object: a shared, low-dimensional structure that anticipates the unobserved late-stimulation response of regulators outside the panel, robustly beats a fair baseline, and remains one pooled object at four times the scale at which it was first validated — with an absolute precision we report honestly as modest.

4. The operator decomposes into gene programs gated by activation state

Section 3 established that the operator has shared, low-dimensional predictive structure. This section asks what that structure *is*: when the operator is decomposed into factors, do those factors correspond to coherent gene programs, and are those programs organized by the biological variable the screen was designed around — the cell’s activation state? We decompose the operator with a CANDECOMP/PARAFAC (CP) factorization, which represents it as a sum of rank-one components, each carrying a regulator loading, a gene loading, and a condition-modulation profile. The condition profile is what makes CP the right tool here: it reads off, for each program, how its transcriptional magnitude varies across resting, early-stimulated (Stim8hr), and late-stimulated (Stim48hr) cells.

Condition gating is the proven result

Two factors are gated by activation state with bootstrap confidence intervals that exclude a flat profile (Figure 3; source `operator_cp_factors_3106.csv`). One peaks in resting cells (factor 1: condition weights 0.91 / 0.39 / 0.15 across Rest / Stim8hr / Stim48hr) and one peaks in early stimulation (factor 6: 0.08 / 0.98 / 0.17). Two further factors are cleanly constitutive — flat across conditions. That both a resting-peaked and a stimulation-peaked program emerge, each with a confidence interval excluding flatness, is the load-bearing claim of this section: the operator’s shared structure is not a single undifferentiated block but resolves into programs whose activity is *modulated by* activation state, in both directions. This is a proven, bootstrap-supported statement about condition identity, and it is what clause (ii) of the thesis claims — nothing more.

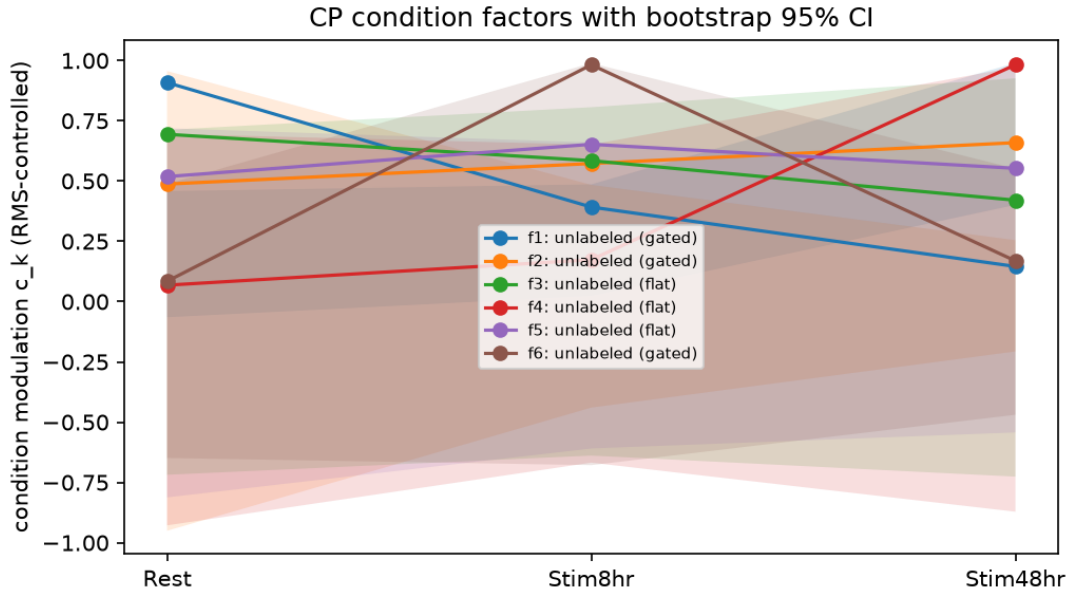


Figure 3. The operator decomposes into gene programs gated by activation state. CANDECOMP/PARAFAC condition-factor weights across Rest / Stim8hr / Stim48hr. Two factors are gated with bootstrap confidence intervals excluding a flat profile — one peaking in resting cells (factor 1) and one in early stimulation (factor 6) — while two further factors are constitutive, evidencing bidirectional gating rather than a single activation program.

We are deliberate about the word “modulated” rather than “activated”: the two gated programs move in opposite directions with stimulation. Reading these as “the activation programs” would miss that one of them is a resting-cell program that stimulation turns *down*. The decomposition recovers bidirectional gating, and the prose should not flatten it into a one-directional activation story.

Naming the programs — non-circular, and honestly thin where it is thin

Assigning a biological name to a factor is a separate, weaker step than proving it is gated, and we hold it to two independent filters. First, non-circularity: a regulator is assigned to a factor by its loading in the CP fit, which is computed with no access to any biological annotation — so recognizing that a factor’s top regulators form a known complex is a genuine test, not a tautology. Second, cis-cleanliness: a regulator can anchor a name only if it survives the cis-off-target gate of Section 2, because a regulator whose apparent effect came from knocking down a neighboring gene is not evidence for anything.

Under both filters, **factor 6 reads as an early-activation program coupled to SAGA chromatin machinery**. Its top regulators include three SAGA subunits — TADA2B, TAF6L, and SUPT20H — but here the cis gate does real work: TADA2B and TAF6L are cis-inflated, collapsing from ranks 2 and 9 to 93 and 130 under a hard cis-off-target exclusion, so the SAGA anchor rests on the one cis-clean subunit present, **SUPT20H**. This makes factor 6’s SAGA signal thinner than its raw subunit list suggests — cis-clean, it is essentially one subunit — and we state it that way rather than citing the full list. Its gene loadings are enriched for early T-cell-activation effector transcripts (leukocyte actin- and TCR-signaling effectors — CORO1A, ARPC1B, GPSM3, LIMD2, MYL12A; empirical “TCR signaling|up” markers, hypergeometric $p = 1.9\text{e-}5$), consistent with the Stim8hr peak. The curated TCR-proximal kinase set (ZAP70/LCK/LAT) shows no overlap with factor 6, as expected — these are downstream read-out genes, not proximal signaling components.

Factor 1 reads as a resting-gated program spanning proteostasis/RNA-processing on the gene side and a multi-module regulator side. Its downstream genes co-load chaperone and RNA-processing transcripts (TCP1, HSPD1, HSPE1, HNRNP family); because the repository carries no curated proteostasis gene set, this is an identity-and-cross-decomposition call, not a hypergeometric p-value, and we label it as such. On the regulator side, factor 1 collects two recognizable modules: the AHR–ARNT xenobiotic-sensing heterodimer, and the immune Rac→WAVE→actin cytoskeletal axis (NCKAP1L/Hem1, WASF2, DOCK2, ARHGAP30). Whether these two modules are one program or two that the rank-6 truncation fuses is a question we can pose but not settle: they co-load at the stable rank and separate only at ranks 8–10, which sit below the decomposition’s stability floor, so the split is suggestive, not decisive — we cannot distinguish two genuine programs emerging from ordinary factor rotation in an unstable regime. We report the co-loading and stop there.

The honest summary of this section is a two-tier one: condition gating is proven with bootstrap CIs; the program *names* are a weaker, filtered overlay, strongest for factor 6’s SUPT20H-anchored SAGA coupling and explicitly tentative for factor 1’s internal structure. The next section tests one of these names against an entirely independent axis of evidence.

5. Universal versus T-cell-specific regulation

The programs named in Section 4 are a within-dataset result: they come from decomposing the CD4⁺ T-cell operator alone. This section tests one of the strongest naming claims — that the immune Rac→WAVE→actin cytoskeletal module is T-cell biology rather than generic cellular machinery — against an entirely independent axis: how the same regulators behave in a different cell type. If a regulator’s perturbation signature is conserved between CD4⁺ T cells and an unrelated lineage, its regulation is likely universal cellular housekeeping; if the signature diverges, the regulation is cell-type-specific. The comparator is the genome-wide Perturb-seq screen of Replogle et al. (2022) in K562, a chronic-myelogenous-leukemia line with no T-cell identity.

Construction and controls

We use Replogle’s pre-aggregated, Z-normalized bulk product — a perturbation $\times$ gene signature matrix, the correct unit for a per-regulator concordance, rather than the 8.8 GB cell-level file. Regulators are collapsed to gene level and non-targeting controls dropped (602 removed), genes are matched by Ensembl identifier, and 6,407 regulators are shared between the two screens. For each shared regulator we compute a per-regulator Pearson concordance between

its Z-scored CD4⁺ and K562 effect profiles over co-measured genes, and convert it to a Z-score against a per-regulator null.

Three controls guard the split. A label-permutation null — permuting the cross-dataset pairing so each CD4⁺ regulator is compared against a random K562 regulator — is centered at zero with a 95th percentile of 0.038, so the observed concordances are not a baseline artifact. A coverage check finds essentially no dependence of concordance on the number of co-measured genes (Spearman -0.010 ; the data are near-complete), so missingness drives nothing. The decisive control is for statistical power: concordance correlates with knockdown strength (Spearman 0.200 against the minimum per-side knockdown rank; source `k562_concordance.csv`, `min_kd_rank` column), meaning a regulator can look “T-specific” simply because it was weakly perturbed in one screen. We therefore do not report the raw split; we lead with the well-powered subset, where both screens knocked the regulator down strongly.

The result, and the inversion

Among well-powered regulators, 508 are universal and 227 are T-cell-specific; restricting further to the donor-reproducible core gives 201 universal and 49 T-specific. Both are conservative counts, and deliberately so — for immune regulators that are lowly expressed in K562, a weak K562 knockdown *is itself* a form of cell-type specificity, so the power filter discards some genuine T-specific signal along with the noise. The T-specific count is a floor, not an estimate.

The biological pattern inverts the naive expectation. One might guess that the activation-coupled program (factor 6) would be the T-cell-specific one and the constitutive machinery universal; the data give the opposite, and it is more coherent for being so. **SAGA chromatin machinery is universal:** SUPT20H — the cis-clean SAGA anchor from Section 4 — is concordant between the two cell types ($Z = +0.186$, well-powered and donor-reproducible). Chromatin/transcription machinery does the same job in any cell, so its conservation is expected. **The immune Rac→WAVE→actin module is T-cell-specific:** its clean anchor is DOCK2, a lymphocyte Rac guanine-nucleotide-exchange factor, which is well-powered and donor-reproducible and shows no cross-cell-type concordance. This is the same module Section 4’s decomposition assigned to factor 1 — so two independent analyses, the within-dataset CP decomposition and the cross-cell-type comparison, converge on the identity of the immune cytoskeletal module as T-cell-specific.

We anchor each arm on a single clean regulator, symmetrically, because that is what the evidence supports. On the universal side, SUPT20H alone survives every filter — TADA2B and TAF6L also score universal but are cis-inflated (Section 4) and not well-powered. On the T-specific side, DOCK2 alone is well-powered; the module’s other members — NCKAP1L (Hem1), AHR, and ARNT — are donor-reproducible and classified T-specific but fall below the well-powered threshold, so their T-specific label rests on the conservative-floor argument rather than on independent power. The strong form of the claim is therefore carried by one clean anchor per arm, SUPT20H and DOCK2, with the surrounding modules consistent but not independently powered — the same evidentiary discipline applied on both sides.

The value of this section is that it converts a within-dataset naming call into a cross-dataset, externally validated distinction: the operator does not merely factor into programs, it separates regulation that is generic to cells from regulation that is specific to the T-cell lineage, and it does so in agreement with an independent screen the decomposition never saw.

6. Discussion

What the operator view buys

The three results share a common source: treating the screen as one object rather than a list. Each is something the ranking or program-extraction views cannot produce on their own. The predictive result (Section 3) is the clearest case — a ranked list says nothing about a regulator outside the panel, whereas the operator’s shared structure predicts the unobserved late-stimulation response of out-of-panel regulators, beating an honest persistence baseline at every rank. The condition-gated programs (Section 4) exist in program-extraction analyses too, but here they emerge as factors of a single object whose condition-modulation profile is read off directly and tested with bootstrap confidence intervals, rather than as separate per-condition decompositions that must then be aligned. And the universal-versus-specific contrast (Section 5) is not a property of the CD4⁺ screen at all until it is set against a second cell type; the operator makes each regulator’s effect profile the unit of comparison, so the contrast becomes a data-built distinction rather than an annotation. The through-line is that shared structure, once made explicit, is usable — for prediction, for condition resolution, and for cross-cell-type comparison — in ways a hit list is not.

Honest limits

We gather the caveats here rather than leave them distributed, because together they define the scope of the claims. First, low-rank is a property of the *predictive subspace*, not the operator: the completion optimum at ~ 7 components describes the transferable structure, while the operator’s own effective rank is roughly an order of magnitude higher. A reader should not take “low-rank operator” as a description of the object — only of the part of it that generalizes. Second, the program names rest on single clean anchors. After the cis-off-target and power filters, the SAGA program is anchored on SUPT20H alone and the T-cell-specific cytoskeletal module on DOCK2 alone; the surrounding subunits and modules are consistent but not independently powered, and we have been explicit about which regulators carry the claim and which ride along on a conservative-floor argument. Third, absolute predictive precision is modest and softens at scale: the out-of-panel model R^2 is 0.06–0.08 across strata, below the 0.154 the smaller pilot reached, so the robust finding is the *advantage* over baseline, not a high absolute accuracy.

Beyond these bounded caveats there is one genuine conceptual gap. We inherit the effect estimates from the screen’s harmonized differential-expression pipeline rather than jointly modeling the raw counts with guide-assignment uncertainty — the territory of calibrated single-cell-CRISPR testing and guided factor models. A fully generative treatment that propagated assignment uncertainty into the operator would be more principled, and we defer it deliberately: it requires the cell-level data (1.8 TB) and the compute that entails, against a design goal of a laptop-reproducible pipeline. Naming this gap is part of the honest accounting, not a claim that our shortcut is free.

Future directions

Three extensions follow naturally. The donor axis can be examined as a subspace-stability question — whether the operator’s leading structure is preserved across donors, measurable as principal angles between per-donor operators (`operator_donor_angles.csv`) — turning donor reproducibility from a per-regulator annotation into a statement about the shared structure itself. The operator’s off-diagonal structure invites a network-deconvolution treatment, which should be benchmarked against the recent single-cell perturbation network-inference benchmark

of Chevalley et al.⁸ rather than asserted, given that field’s finding that inference methods often fail to beat simple baselines. And the low-power regulator tail, excluded here by the cell-count floor, could be recovered with a precision-weighted (lfcSE) sub-floor build if those specific regulators — rather than the well-powered structure — became the target of a follow-up.

The contribution of this work is not a new estimator or a new master regulator, but a way of looking: the perturbation screen as a single operator whose geometry is predictive, condition-resolved, and cell-type-discriminating — recovered with explicit statistical rigor from a 1.8 TB atlas, on a laptop.

7. Methods

Dataset and access

We analyze the genome-scale CRISPRi Perturb-seq screen in primary human CD4⁺ T cells (four donors; three activation states: resting, Stim8hr, Stim48hr). The full atlas is ~1.8 TB; all analyses in this paper run without loading the cell-level data. Instead we use the published supplementary differential-expression tables (~15 MB), a remote per-condition z-score slice fetched on demand (~0.1 GB, cached locally), and, for the cross-cell-type comparison, the Replogle et al. (2022) pre-aggregated K562 bulk product (~375 MB). The entire pipeline is therefore reproducible on a laptop; no step requires the cell-level `.h5ad/.h5mu` files.

Effect estimates and the power confound

Per-regulator effect profiles are the calibrated differential-expression estimates from the screen’s harmonized pipeline; we do not re-derive them. Because raw log-fold-change magnitude is confounded with statistical power — regulators assayed in more cells show systematically larger apparent effects (Spearman ρ between per-regulator effect-vector norm and cell count ≈ -0.69 in raw space) — all operator analyses are performed in a pooled z-score space, in which each entry is standardized at the layer level across the full regulator population rather than per condition.

Operator tensor construction

The operator is assembled as a regulator $\times$ gene $\times$ condition tensor (`build_operator_tensor.py`). Genes are the top-variance panel (2,000 genes by default). Regulators are selected by a per-condition cell-count floor (retaining regulators above the median), which yields 3,106 regulators for the escalated analysis and 800 for the validation pilot. After the z-score fetch, a confound guard re-measures the rank correlation between per-regulator slab norm and cell count on the panel genes and refuses to cache the tensor if $|\rho| \geq 0.15$ (fail-closed); the escalated tensor passes at $\rho = -0.006$, and a power gate on the leading singular subspace holds at $\max |\rho| = 0.049$.

CP decomposition

The tensor is decomposed with a masked CANDECOMP/PARAFAC factorization (`decompose_operator_cp.py`; `tensorly parafac` with a missingness mask, `n_iter_max = 400`, `n_init = 10`, `random_state = 0`). Before fitting, condition slabs are RMS scale-controlled so that a single high-variance condition cannot dominate the fit. Rank is selected by split-half stability: for each candidate rank we compute the matched-factor cosine between decompositions of two random halves (subsample 400), and take the largest rank whose

stability exceeds 0.70. Each factor carries a regulator loading, a gene loading, and a three-entry condition-modulation profile; condition gating is assessed by bootstrap (100 resamples over conditions), and a factor is called gated only if the bootstrap confidence interval on its modulation profile excludes a flat profile. Degenerate factors — those with a maximum cross-factor cofactor cosine near 1 (≈ 0.97) — are excluded before interpretation. Factor loadings are varimax-rotated (`gamma = 1.0`, convergence on rotation change, `tol = 1e-6`).

Out-of-panel completion

Predictive structure is assessed by low-rank matrix completion of a held-out condition fiber (`operator_completion.py`, kernels in `scripts/_opkernels.py`). We hold out the late-stimulation (Stim48hr) entries of regulators outside the original variance-selected panel (a seed-0 20% sample; `--holdout 0.2 --seed 0`), fit a low-rank model to the remaining entries, and predict the held-out fiber. Standardization is computed on the training entries only (`train_test_standardize`, centering each gene on its train-mask values) to prevent leakage. Completion uses soft-impute with a truncated SVD (`soft_impute`, `n_iter = 100`, `tol = 1e-4`); the truncated solver (`scipy.sparse.linalg.svds`) is numerically identical to a full-SVD soft-impute to machine precision (a regression test asserts agreement to $\sim 1e-13$) and is roughly an order of magnitude faster. The baseline is persistence — predicting the late-stimulation response equals the early-stimulation one. We report the model–persistence margin per rank, and stratify the held-out regulators by knockdown strength (`operator_completion_stratified.py`), with the stratum thresholds hardcoded and documented in that script: a median split on effect strength, and a panel-matched stratum (regulators at least as strong as the median in-panel regulator).

K562 cross-cell-type concordance

Cell-type specificity is assessed against the Replogle et al. (2022) genome-wide K562 Perturb-seq screen, using its pre-aggregated, Z-normalized perturbation $\times$ gene bulk product (`K562_gwps_normalized_bulk_01.h5ad`). Regulators are collapsed to gene level with an unweighted mean over guides/promoters, non-targeting controls are dropped (602 removed), and genes are matched by Ensembl identifier, yielding 6,407 shared regulators. For each shared regulator we compute a Pearson concordance between its Z-scored CD4⁺ and K562 effect profiles over co-measured genes (no imputation of non-measured entries), converted to a per-regulator Z-score. Three controls are applied: a label-permutation null (permuting the cross-dataset pairing; 95th percentile 0.038); a coverage check (Spearman between concordance and number of co-measured genes, -0.010); and a power control (Spearman between concordance and the minimum per-side knockdown rank, 0.200 via the `min_kd_rank` column). Because concordance depends on power, we lead with the well-powered subset (both screens knocking the regulator down strongly) and report the donor-reproducible core as a conservative floor. Regulators are classified universal / T-cell-specific / intermediate on the sign and magnitude of their concordance Z-score.

Reproducibility

The full operator pipeline runs from `make operator` (tensor $\rightarrow$ SVD $\rightarrow$ CP $\rightarrow$ completion $\rightarrow$ donor angles). Numerical kernels are isolated in `scripts/_opkernels.py` and covered by a unit-test suite (17 tests, `tests/test_opkernels.py`), including the svds/full-SVD equivalence regression test. Random seeds are fixed (`random_state = 0`, `--seed 0`) throughout. All result tables cited in the text are written to `docs/tables/` with a `_3106` suffix for the escalated anal-

ysis, and the pilot-scale tables are retained alongside them so the pilot↔escalation comparison is itself reproducible from disk.

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